

Where each method injects the signal — and what it bought

channel into the association

source of the signal				
	Box size (geometry)	Detection confidence	Localization quality	Detector σ (NMS / DFL)
Cost: additive				<i>ours</i> −3.75
Cost: multiplicative		ByteTrack impl (no rationale)	LG-Track eq.1, 3	<i>ours</i> −1.01
Distance function	NWD $\Sigma = \text{diag}(w^2/4, h^2/4)$ (detection); <i>ours</i> −4.98			<i>ours (Wasserstein)</i> −8.90; loses to size by 3.92
Kalman R	UCMCTrack $R = (\sigma_m w, \sigma_m h)$	NSA Kalman $\hat{R} = (1 - c)R$		UTrack −0.62 UncTrack −0.1
Gating / box expansion	<i>ours (control)</i> beats σ by 3.81			UncTrack +2.2 (expansion) <i>ours</i> : σ -driven −4.33
Association ordering			LG-Track 4-level cascade	UncTrack entropy +0.2
Matching threshold				<i>ours</i> −0.21 (<i>oracle</i> +0.89)

italic = this work

gain reported loss measured neutral / baseline mixed (different quantities)